

Standard deviations do not exist outside the Gaussian, or if they do exist they do not matter and do not explain much. But it gets worse. The Gaussian family (which includes various friends and relatives, such as the Poisson law) are the only class of distributions that the standard deviation (and the average) is sufficient to describe. You need nothing else. The bell curve satisfies the reductionism of the deluded.

There are other notions that have little or no significance outside of the Gaussian: *correlation* and, worse, *regression*. Yet they are deeply ingrained in our methods; it is hard to have a business conversation without hearing the word *correlation*.

To see how meaningless correlation can be outside of Mediocristan, take a historical series involving two variables that are patently from Extremistan, such as the bond and the stock markets, or two securities prices, or two variables like, say, changes in book sales of children's books in the United States, and fertilizer production in China; or real-estate prices in New York City and returns of the Mongolian stock market. Measure correlation between the pairs of variables in different subperiods, say, for 1994, 1995, 1996, etc. The correlation measure will be likely to exhibit severe instability; it will depend on the period for which it was computed. Yet people talk about correlation as if it were something real, making it tangible, investing it with a physical property, reifying it.

The same illusion of concreteness affects what we call "standard" deviations. Take any series of historical prices or values. Break it up into subsegments and measure its "standard" deviation. Surprised? Every sample will yield a different "standard" deviation. Then why do people talk about standard deviations? Go figure.

Note here that, as with the narrative fallacy, when you look at past data and compute one single correlation or standard deviation, you do not notice such instability.

How to Cause Catastrophes

If you use the term *statistically significant*, beware of the illusions of certainties. Odds are that someone has looked at his observation errors and assumed that they were Gaussian, which necessitates a Gaussian context, namely, Mediocristan, for it to be acceptable.

To show how endemic the problem of misusing the Gaussian is, and how dangerous it can be, consider a (dull) book called *Catastrophe* by Judge Richard Posner, a prolific writer. Posner bemoans civil servants' misunderstandings of randomness and recommends, among other things, that